

Dr. Utkarsh Jadli

Electronics & Embedded Systems Engineer

Brisbane, QLD, Australia

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PROFESSIONAL SUMMARY

Electronics and Embedded System Engineer with a PhD in Power Semiconductor Devices and 4.5 years delivering certified hardware systems for safety-critical, harsh-environment field applications. Experienced across the full hardware lifecycle from requirements definition and architecture through PCB design, embedded firmware, verification and validation, and live deployment. Strong background in precision analog front-end design, mixed-signal systems, multi-sensor integration, EMI/EMC testing, wiring and telemetry systems, and environmental test campaigns. Certified hardware designed under IEC 60079 and IEC 80079 standards, with formal DFMEA, requirements traceability, and independent certification authority review. Applying the same engineering rigour — margin analysis, fault tolerance, and first-time-correct design — to electronic systems.

Portfolio: utkarshjadli-bit.github.io

CORE COMPETENCIES

Safety-Critical Hardware Design | IEC 60079/80079 Certified | DFMEA | Requirements Traceability
Mixed-Signal PCB Design (Altium) | Wiring & Cable Systems | Low-Power Battery Systems
EMI/EMC Testing | Environmental Testing | Thermal & Vibration Campaigns
STM32, ESP32, ARM Cortex-M, Raspberry Pi | CAN bus, I2C, SPI, UART, HART, BLE/Wi-Fi
Verification & Validation (V&V) — Unit, Subsystem & System Level | IEC 60601/IEC 62366 Familiarity
Root Cause Investigation | COTS Evaluation | Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Sigra Pty Ltd – Brisbane, Australia

Nov 2021 – Feb 2026

Electronics Engineer — Safety-Critical Instrumentation

Verification, Validation & Certification

- Planned and executed full V&V campaigns for a 24-bit strain acquisition system: beam-bending validation against analytical models, 67-hour thermal drift characterisation across a 39.6°C temperature swing, and mechanical testing using UCS equipment.
- Resolved 11 technical queries from an independent certification authority during certification of an intrinsically safe battery management system, covering fault energy calculations, component derating, and worst-case failure scenarios.
- Developed test procedures, formal acceptance criteria, and requirements traceability documentation to close certification requirements under IEC 60079 and IEC 80079.
- Performed DFMEA, worst-case capacitance analysis, and component derating analysis as part of safety case development for hazardous area hardware.
- Conducted EMI/EMC verification and signal integrity testing on mixed-signal PCBs using oscilloscopes, logic analysers, protocol analysers, and spectrum analysis to validate performance under interference conditions.

Hardware Design, Wiring & Systems Engineering

- Architected a 24-bit strain acquisition platform achieving 17.8-bit ENOB and 1.83 $\mu\epsilon$ RMS noise within a constrained 27 mm cylindrical form factor, including low-power battery management and passive Hall-effect power gating for zero-intervention downhole operation.
- Designed wiring harnesses and cable assemblies for multi-sensor downhole instruments over 1.4 km of armoured cable, including connector selection and continuity/insulation testing.
- Developed IMU-based borehole survey system achieving $\pm 0.33^\circ$ azimuth, $\pm 0.11^\circ$ inclination, and $\pm 0.25^\circ$ toolface accuracy; validated HART telemetry across three independent tool builds.
- Designed low-noise analog front-end circuitry including instrumentation amplification, filtering, ADC interfacing, and sensor excitation for high-accuracy measurement under harsh field conditions.
- Designed and debugged multi-layer PCBs in Altium Designer with emphasis on mixed-signal layout, grounding strategy, and EMI mitigation.

Embedded Firmware, Integration & Field Deployment

- Developed embedded firmware in C for STM32 and ESP32 microcontrollers: sensor interfacing (CAN bus, I2C, SPI, UART), data acquisition, calibration routines, and HART/RS-485/Modbus telemetry.

- Integrated multi-sensor subsystems (strain gauges, IMUs, thermistors, Hall-effect, ToF) from schematic through field deployment; used oscilloscopes, logic and protocol analysers, DMM, and LCR meters for board bring-up and fault diagnosis.
- Leveraged ESP32 Wi-Fi and Bluetooth/BLE connectivity for IoT sensor nodes; used Raspberry Pi and Arduino platforms for rapid prototyping and test infrastructure.
- Delivered a \$250k+ ACARP engineering project on schedule during global semiconductor shortages, managing COTS component qualification, evaluation, and adaptive PCB redesign.
- Conducted on-site root cause investigation and troubleshooting across active mining sites under operational time constraints, maintaining system uptime under pressure.

RESEARCH & ENGINEERING EXPERIENCE

Griffith University – [Brisbane, Australia](#)

2018 – 2021

PhD Researcher — Power Semiconductor Device Modelling & Characterisation

- Characterised Si MOSFETs, SiC MOSFETs, and GaN HEMTs via I-V and C-V measurements across temperature using Keysight B1506A and HP 4156C precision parameter analysers. SiC and GaN are increasingly adopted in aerospace and space power conversion for radiation tolerance, high-temperature operation, and switching efficiency.
- Developed SPICE-based compact models including parasitic capacitance extraction and double-pulse test validation for dynamic switching behaviour.

QMNC, Griffith University – [Brisbane, Australia](#)

2019 – 2021

Research Assistant / Fellow

- Designed and debugged PCBs in Altium for laboratory sensor systems and automated measurement setups. *Designed flexible EMG sensor PCBs for biomedical signal acquisition (Altium Designer).*

Griffith University – [Brisbane, Australia](#)

Jul 2018 – Nov 2021

Sessional Staff — Laboratory Instructor

Led laboratory sessions for Semiconductor Devices and Circuits, Integrated Circuit Design, Practical Electronics, and Data and Computer Communications. Nominated twice for Sessional Staff Award.

EDUCATION

Ph.D. — Power Semiconductor Devices, Griffith University (Brisbane)

2018 – 2021

M.Tech — Control Engineering, Graphic Era University (Dehradun)

2015 – 2017

B.Tech — Electrical & Electronic Engineering, Graphic Era University (Dehradun)

2011 – 2015

TECHNICAL SKILLS

Hardware Design: Mixed-Signal Circuit Design, Precision Analog Front Ends, Sensor Excitation & Conditioning, Low-Power Battery Systems, Wiring & Cable Assemblies, EMI Mitigation, Multi-layer PCB Design (Altium), DFM/DFT

Embedded Systems: STM32, ESP32, ARM Cortex-M, Raspberry Pi, Arduino, Embedded C, CAN bus, I2C, SPI, UART, Bluetooth/BLE, Wi-Fi (ESP32), Data Acquisition, Calibration Workflows

Test & Measurement: Oscilloscopes, Logic Analysers, Protocol Analysers, DMM, LCR Meters, Spectrum Analysers, EMI/EMC Testing, Environmental Testing, Thermal Analysis, Vibration & Integration Testing, Root Cause Investigation

Communication Protocols: CAN bus, UART, RS-485, Modbus RTU, HART, I2C, SPI, Bluetooth/BLE, Wi-Fi

Simulation & Modelling: LTSPICE/SPICE, MATLAB/Simulink, Python (NumPy, SciPy)

PUBLICATIONS & PROFESSIONAL MEMBERSHIP

Published research on power semiconductor device modelling and characterisation (SiC/GaN) and power electronics simulation. 17 papers, 400+ citations.

Google Scholar: scholar.google.com.au/citations?user=kmwQvPQAAAAJ

Member: IEEE Power Electronics Society | IEEE Electron Device Society